ACADEMIC PROJECT PROPOSAL & ABSTRACT

EduTrack: Secure Full-Stack Student & Department Management Platform

Full-Stack Web Application Built with Spring Boot 3, React 19, and MySQL

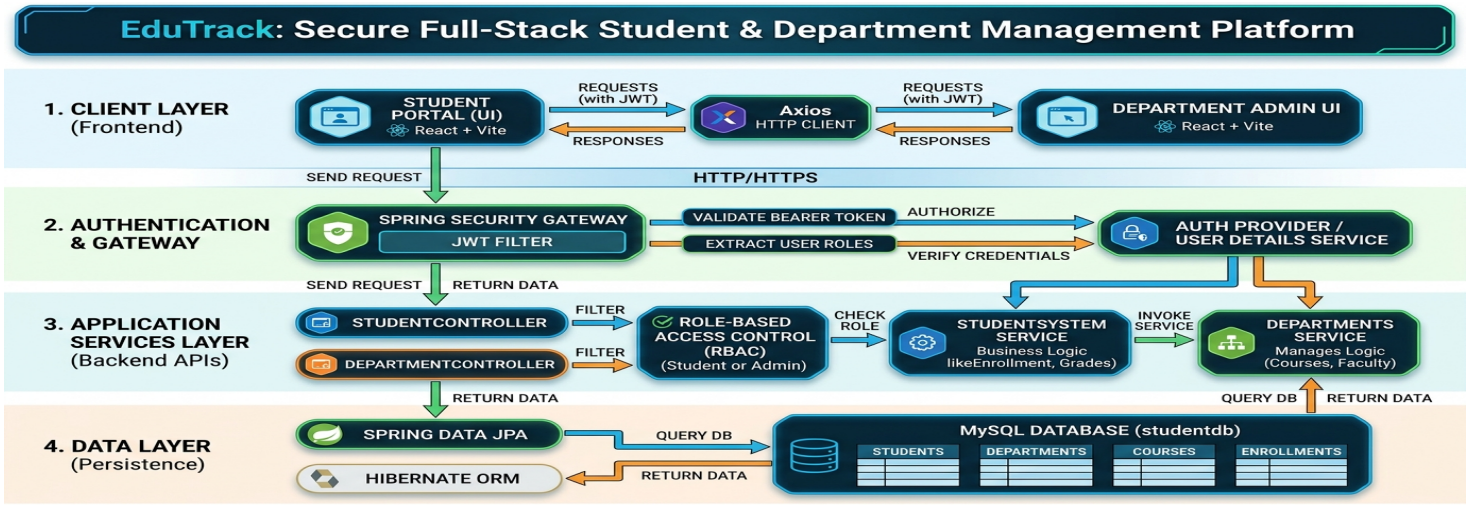
1. Project Abstract

EduTrack is an enterprise-grade academic administration platform engineered with a decoupled full-stack architecture. The platform addresses fragmented university departmental records by providing centralized student and academic department lifecycle management. Built with a reactive Single Page Application (SPA) frontend in React 19 and a secure Spring Boot 3 RESTful API backend, EduTrack implements stateless authentication via JSON Web Tokens (JWT) and BCrypt password hashing. Key capabilities include role-based dashboards for System Administrators and Students, departmental enrollment tracking, live search and filtering, and full CRUD operations mapped via Spring Data JPA and Hibernate ORM.

2. Technology Stack & Architectural Components

Tier	Technology	Role & Responsibilities
Frontend (Client)	React 19, Vite, Axios, React Router	Single-page UI, client-side routing, JWT interceptors, modal dialogs
Backend (API)	Spring Boot 3.4.3 (Java 21)	REST controllers, service business logic, DTO mapping & validation
Security & Auth	Spring Security 6, JJWT, BCrypt	Stateless JWT token validation filter & Role-Based Access Control (RBAC)
Persistence & DB	Spring Data JPA, Hibernate, MySQL	Relational schema persistence (studentdb) with H2 embedded fallback

3. Architecture & Data Flowchart



4. Functional Modules

- **Authentication & Security:** JWT bearer token generation, BCrypt password hashing, role segregation (ADMIN vs STUDENT).
- **Department Administration:** Full CRUD for academic faculties, unique course codes (CSE, IT, ECE), and live headcount calculation.
- **Student Information System:** Multi-parameter search (name/email), departmental filtering, and dynamic profile editing.
- **Analytics Dashboard:** Real-time KPI summary metrics displaying total enrolled students, academic departments, and role permissions.